

# Chapter 1, Section 3

James Lee

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2. A morphism whose underlying map on the topological space is a homeomorphism need not be an isomorphism.

- (a) For example, let  $\varphi : \mathbb{A}^1 \rightarrow \mathbb{A}^2$  be defined by  $t \mapsto (t^2, t^3)$ . Show that  $\varphi$  defines a bijective bicontinuous morphism of  $\mathbb{A}^1$  onto the curve  $y^2 = x^3$ , but that  $\varphi$  is not an isomorphism.
- (b) For another example, let the characteristic of the base field  $k$  be  $p > 0$ , and define a map  $\varphi : \mathbb{A}^1 \rightarrow \mathbb{A}^1$  by  $t \mapsto t^p$ . Show that  $\varphi$  is bijective and bicontinuous but not an isomorphism. This is called the *Frobenius morphism*.

*Proof.*

- (a) Let  $Y$  be the curve  $y^2 = x^3$ . Then  $\varphi$  clearly maps into  $Y$  and is injective since

$$(t^2, t^3) = (u^2, u^3) \implies tu^2 = t^2u \implies t = u.$$

If  $P = (a_1, a_2) \in Y$ , then clearly  $\varphi(\sqrt[3]{a_2}) = P$  since  $k$  is algebraically closed, hence  $\varphi$  is bijective onto  $Y$ . To show  $\varphi$  is bicontinuous it is enough to show  $\varphi$  and  $\varphi^{-1}$  of closed sets in  $\mathbb{A}^1$  and  $Y$ , respectively, is closed. A closed set in either  $\mathbb{A}^1$  or  $Y$  is a finite set of points, so by the bijectivity of  $\varphi$ , it is also bicontinuous.

To show it is not an isomorphism, it suffices to show the coordinate rings of the affine line  $\mathbb{A}^1$  and the curve  $Y$  are not isomorphic. Indeed,  $A(\mathbb{A}^1) = k[x]$  and  $A(Y) = k[x, y]/(y^2 - x^3)$ , and while  $k[x]$  is factorial,  $A(Y)$  is not since  $y^2 = x^3$ .

- (b) It is bijective and bicontinuous for the same reason in (a). If  $\varphi$  is an isomorphism, then it must induce an automorphism  $\varphi^* : k[x] \rightarrow k[x]$  of the affine coordinate ring  $k[x]$  of  $\mathbb{A}^1$ . However,  $\varphi^*$  is defined by  $x \mapsto x^p$ , which is not even surjective.

□

- 3. (a) Let  $\varphi : X \rightarrow Y$  be a morphism. Then for each  $P \in X$ ,  $\varphi$  induces a homomorphism of local rings  $\varphi_P^* : \mathcal{O}_{\varphi(P), Y} \rightarrow \mathcal{O}_{P, X}$ .
- (b) Show that a morphism  $\varphi$  is an isomorphism if and only if  $\varphi$  is a homeomorphism, and that the induced map  $\varphi_P^*$  on local rings is an isomorphism for all  $P \in X$ .
- (c) Show that if  $\varphi(X)$  is dense in  $Y$ , then the map  $\varphi_P^*$  is *injective* for all  $P \in X$ .

*Proof.*

- (a) We have already seen that  $\varphi$  induces a homomorphism  $\varphi^* : \mathcal{O}(Y) \rightarrow \mathcal{O}(X)$ . By (2.3), there exists an open affine variety  $V$  in  $Y$  containing  $\varphi(P)$ . Then  $U = \varphi^{-1}(V)$  contains  $P$ , so  $\mathcal{O}_{P, X} \simeq \mathcal{O}_{P, U}$  and  $\mathcal{O}_{\varphi(P), Y} \simeq \mathcal{O}_{\varphi(P), V}$ . By definition  $f$  is a regular function defined in some open neighborhood of  $\varphi(P)$  in  $V$  such that  $f(\varphi(P))$  is nonzero if and only if  $\varphi^* f$  is nonzero at  $P$ . Thus,  $\varphi^* f$  is a unit in  $\mathcal{O}_{P, U}$ , so by the universal property of localization there exists a unique ring homeomorphism  $\varphi_P^* : \mathcal{O}_{\varphi(P), Y} \rightarrow \mathcal{O}_{P, X}$ .
- (b) The only if direction is obvious. Conversely, if  $\varphi_P^*$  on local rings is an isomorphism for all  $P \in X$ , then since isomorphism is a local property (A.M. p. 40), we have an isomorphism  $\varphi^* : \mathcal{O}_V \rightarrow \mathcal{O}_U$ , where  $V$  is any affine variety in  $Y$  and  $U = \varphi^{-1}(V)$ . Hence,  $\varphi : U \rightarrow V$  is an isomorphism.
- (c) We want to show for any  $P \in X$  that if  $f, g$  are elements of  $\mathcal{O}_{\varphi(P), Y}$  such that  $\varphi_P^* f = \varphi_P^* g$  on some open neighborhood  $U$  of  $P$ , then  $f = g$  in some open neighborhood  $V$  of  $\varphi(P)$ . Let  $V$  be an open neighborhood of  $\varphi(P)$  with both  $f$  and  $g$  are defined so that  $\varphi_P^* f$  and  $\varphi_P^* g$  are defined on  $U = \varphi^{-1}(V)$ . Then,  $\varphi(U) = V \cap \varphi(X)$  is a closed and dense subset of  $V$  such that  $f = g$ , hence  $f = g$  on  $V$ .

□

4. Show that the  $d$ -uple embedding of  $\mathbb{P}^n$  (I, Ex. 2.12) is an isomorphism onto its image.

*Proof.* Let  $\rho : \mathbb{P}^n \rightarrow Y \subseteq \mathbb{P}^N$  be the  $d$ -uple embedding of  $\mathbb{P}^n$  with  $N = \binom{n+d}{d} - 1$  and image restricted so that  $\rho$  is a homeomorphism. By Exercise 2, to show  $\rho$  is an isomorphism it suffices to show it is an isomorphism locally. Let  $P = (a_0, \dots, a_n) \in \mathbb{P}^n$  and  $\rho(P) = (b_0, \dots, b_N) \in Y$  and assume  $a_0 \neq 0$ . By (3.4), we have the following identifications of local rings

$$\mathcal{O}_{P, \mathbb{P}^n} = S(\mathbb{P}^n)_{(\mathfrak{m}_P)}, \quad \mathcal{O}_{\rho(P), Y} = S(Y)_{(\mathfrak{m}_{\rho(P)})},$$

where  $\mathfrak{m}_P$  (respectively,  $\mathfrak{m}_{\rho(P)}$ ) is the ideal generated by the set of homogenous  $f \in S(\mathbb{P}^n)$  (respectively,  $f \in S(Y)$ ) such that  $f(P) = 0$  (respectively,  $f(\rho(P)) = 0$ ). We want to show the induced map  $\rho_P^* : \mathcal{O}_{\rho(P), Y} \rightarrow \mathcal{O}_{P, \mathbb{P}^n}$  is an isomorphism. By the identification above, the map  $\rho_P^*$  can be defined by the canonical projection map  $\bar{\theta} : k[y_0, \dots, y_N]/\mathfrak{a} \rightarrow k[x_0, \dots, x_n]$  induced by  $\theta$  in Exercise 2.12 as

$$\rho_P^*\left(\frac{f}{g}\right) = \frac{\bar{\theta}(f)}{\bar{\theta}(g)},$$

where  $f$  and  $g$  are homogenous elements of same degree in  $S(Y)$  such that  $g(\rho(P)) \neq 0$  (since we are only concerned with homogenous elements of degree 0 in  $S(Y)_{\mathfrak{m}_{\rho(P)}}$ ). Clearly  $\rho_P^*$  is injective since  $\bar{\theta}$  is injective, and  $\rho_P^*$  maps  $S(Y)_{(\mathfrak{m}_{\rho(P)})}$  into  $S(\mathbb{P}^n)_{(\mathfrak{m}_P)}$  since  $\theta(y_i)$  is of the same degree for all  $0 \leq i \leq N$  and  $\mathfrak{a}$  is a homogenous ideal. To show it is surjective, suppose  $h/k \in \mathcal{O}_{P, \mathbb{P}^n}$  with  $k(P) \neq 0$  and  $h, k$  both homogenous of degree  $e$ . We can assume  $e$  is a multiple  $d$  since  $x_0$  is a unit in  $\mathcal{O}_{P, \mathbb{P}^n}$ , and we can multiply  $h/k$  by a sufficient power of  $x_0/x_0$  to obtain the desired degrees in the numerator and denominator. Then, each term in  $h$  and  $k$  is the product of monomials of degree  $d$ , and since  $\bar{\theta}$  is surjective onto such elements,  $\rho_P^*$  is surjective. Hence,  $\rho_P^*$  is an isomorphism. □

6. There are quasi-affine varieties which are not affine. For example, show that  $X = \mathbb{A}^2 - \{(0, 0)\}$  is not affine.

*Proof.* The ideal  $I(X)$  consists of all polynomials in  $k[x, y]$  that vanish at  $X$ . If  $f \in I(X)$ , then viewing  $f$  as a regular function on  $\mathbb{A}^2$ ,  $f^{-1}(0)$  is a closed and dense subset of  $\mathbb{A}^2$ , hence it equals to  $\mathbb{A}^2$ , hence  $I(X) = (0)$ . Thus, if  $X$  is affine, then

$$\mathcal{O}(X) = k[x, y]/I(X) \simeq k[x, y] = \mathcal{O}(\mathbb{A}^2),$$

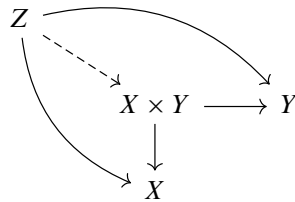
so by (3.5) and (3.7)  $X \cong \mathbb{A}^2$  under the inclusion map, which is ridiculous. □

12. If  $P$  is a point on a variety  $X$ , then  $\dim \mathcal{O}_P = \dim X$ .

*Proof.* We can find an open affine variety  $Y$  in  $X$  containing  $P$ , and since local rings behave well under open subsets, i.e.  $\mathcal{O}_{P, Y} \simeq \mathcal{O}_{P, X}$ , we can reduce to the affine case. Then, the statement follows from (3.2). □

15. *Products of Affine Varieties.* Let  $X \subseteq \mathbb{A}^n$  and  $Y \subseteq \mathbb{A}^m$  be affine varieties.

- Show that  $X \times Y \subseteq \mathbb{A}^{n+m}$  with its induced topology is irreducible. The affine variety  $X \times Y$  is called the *product* of  $X$  and  $Y$ . Note that its topology is in general not equal to the product topology.
- Show that  $A(X \times Y) \simeq A(X) \otimes_k A(Y)$ .
- Show that  $X \times Y$  is a product in the category of varieties.
- Show that  $\dim X \times Y = \dim X + \dim Y$ .



*Proof.*

- (a) Suppose that  $X \times Y$  is a union of two closed subsets  $Z_1 \cup Z_2$ . Let  $X_i = \{x \in X \mid x \times Y \subseteq Z_i\}$ ,  $i = 1, 2$  and let  $\pi : \mathbb{A}^{n+m} \rightarrow \mathbb{A}^m$  be the projection map. For any  $x \in X$  since  $x \times Y \subseteq Z_1 \cup Z_2$ , we can write  $x \times Y = (x \times Y \cap Z_1) \cup (x \times Y \cap Z_2)$ , so let  $C_i = x \times Y \cap Z_i$ , then

$$Y = \pi(x \times Y) = \pi(C_1 \cup C_2) = \pi(C_1) \cup \pi(C_2).$$

We can write  $Z_i = Z(f_1, \dots, f_r)$ ,  $i = 1, 2$ , for some  $f_j \in A(\mathbb{A}^{n+m})$ , then we see that

$$\pi(C_i) = \bigcap_{j=1}^r Z(f_j(x, y_1, \dots, y_m)),$$

where we can view  $f_j(x, y_1, \dots, y_m)$  as an element in  $A(\mathbb{A}^m)$ , which shows  $\pi_2(C_i)$  is a closed set in  $\mathbb{A}^m$ . Since  $Y$  is irreducible, we have  $\pi(C_2) = \emptyset$ , say, hence  $Y = \pi(C_1)$ , hence  $x \times Y = x \times Y \cap Z_1$ , hence  $x \times Y \subseteq Z_1$ . In particular, we have  $X = X_1 \cup X_2$ , and we have shown  $Z_i = X_i \times Y$ . We can repeat the argument for  $Y_i = \{y \in Y \mid X \times Y \subseteq Z_i\}$ ,  $i = 1, 2$  so that  $Z_i = X \times Y_i$ . This is only possible if  $X = X_1$  and  $Y = Y_1$ , say, hence  $Z_1 = X \times Y$ .

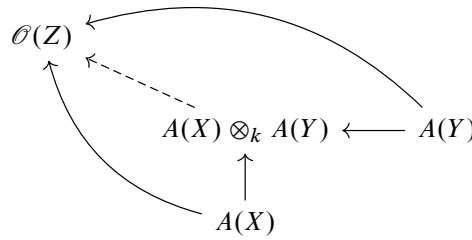
- (b) We clearly have  $A(\mathbb{A}^{n+m}) \simeq A(\mathbb{A}^n) \otimes_k A(\mathbb{A}^m)$ , where  $A(\mathbb{A}^n) = k[x_1, \dots, x_n]$  and  $A(\mathbb{A}^m) = k[y_1, \dots, y_m]$ . Then, we have

$$I(X \times Y) = I(X) \otimes_k A(\mathbb{A}^m) + A(\mathbb{A}^n) \otimes_k I(Y),$$

hence we have the isomorphism

$$\begin{aligned} A(X \times Y) &\cong \frac{A(\mathbb{A}^n) \otimes_k A(\mathbb{A}^m)}{I(X \times Y)} \\ &\cong \frac{A(\mathbb{A}^n) \otimes_k A(\mathbb{A}^m)}{I(X) \otimes_k A(\mathbb{A}^m) + A(\mathbb{A}^n) \otimes_k I(Y)} \\ &\cong \frac{A(\mathbb{A}^n)}{I(X)} \otimes_k \frac{A(\mathbb{A}^m)}{I(Y)} \\ &\cong A(X) \otimes_k A(Y). \end{aligned}$$

- (c) Suppose we have morphisms  $Z \rightarrow X$  and  $Z \rightarrow Y$ . This induces  $k$ -algebra homomorphisms  $A(X) \rightarrow \mathcal{O}(Z)$  and  $A(Y) \rightarrow \mathcal{O}(Z)$ , which defines a  $k$ -bilinear map  $A(X) \times A(Y) \rightarrow \mathcal{O}(Z)$ . By the universal property of tensor products, we have a unique  $k$ -algebra homomorphism  $A(X) \otimes_k A(Y) \rightarrow \mathcal{O}(Z)$ , hence we have a unique morphism  $W \rightarrow X \times Y$ .



- (d) If  $t_1, \dots, t_n$  and  $u_1, \dots, u_m$  are coordinates for  $X$  and  $Y$ , respectively, then each  $t_i, u_j$  are algebraically independent in  $A(X \times Y)$ , that is the quotient field of  $A(X \times Y)$  has transcendence degree  $n + m$ , hence  $\dim X \times Y = \dim X + \dim Y$ .

□

**17. Normal Varieties.** A variety  $Y$  is *normal at a point*  $P \in Y$  if  $\mathcal{O}_P$  is integrally closed ring.  $Y$  is *normal* if it is normal at every point.

- Show that every conic in  $\mathbb{P}^2$  is normal.
- Show that the quadric surfaces  $Q_1, Q_2$  in  $\mathbb{P}^3$  given by equations  $Q_1 : xy = zw$ ;  $Q_2 : xy = z^2$  are normal.
- Show that the cuspidal cubic  $y^2 = x^3$  in  $\mathbb{A}^2$  is not normal.
- If  $Y$  is affine, then  $Y$  is normal  $\iff A(Y)$  is integrally closed.
- Let  $Y$  be an affine variety. Show that there is a normal affine variety  $\hat{Y}$ , and a morphism  $\pi : \hat{Y} \rightarrow Y$ , with the property that whenever  $Z$  is normal variety, and  $\varphi : Z \rightarrow Y$  is a *dominant* morphism, then there is a unique morphism  $\theta : Z \rightarrow \hat{Y}$  such that  $\varphi = \pi \circ \theta$ .  $\hat{Y}$  is called the *normalization* of  $Y$ .

*Proof.*

- (a) A conic in  $\mathbb{P}^2$  can be covered by open affine varieties isomorphic to a conic in  $\mathbb{A}^2$ . Thus, it suffices to show all irreducible plane algebraic curves of degree 2 are normal, that is we want to show an affine variety  $X$  in  $\mathbb{A}^2$  defined by the zero set of an irreducible quadratic polynomial  $f$  in  $k[x, y]$  is normal. Since being integrally closed is a local property, it suffices to show the affine coordinate ring  $A(X) = k[x, y]/(f)$  is integrally closed. We first prove (Ex. 1.1c), which states  $A(X)$  is isomorphic to either  $R = k[x, y]/(y - x^2)$  or  $S = k[x, y]/(xy - 1) \simeq k[x, \frac{1}{x}]$ . For simplicity assume  $\text{char } k \neq 2$ . Any quadratic polynomial can be written as

$$f(x, y) = Ax^2 + Bxy + Cy^2 + Dx + Ey + F$$

We claim the following: if  $B^2 - 4AC = 0$ , then  $A(X) \simeq R$ , otherwise  $A(X) \simeq S$ . We proceed by showing that  $X$  can be transformed to a variety of the form  $y = x^2$  or  $xy = 1$  using some affine transformations.

Suppose  $B^2 - 4AC = 0$ , then assume  $C \neq 0$  (if  $C = 0$ , then  $B = 0$ , so we already have an equation of the desired form up to some translation and stretching), so we have

$$f(x, y) = (\sqrt{A}x + \sqrt{C}y)^2 + Dx + Ey + F,$$

since  $k$  is algebraically closed. We can apply the following affine transformation

$$\begin{aligned} x &\mapsto -\sqrt{C}y \\ y &\mapsto -\sqrt{C}x + \sqrt{A}y \end{aligned}$$

which is indeed an affine transformation since it has determinant  $C \neq 0$ , to obtain

$$f'(x, y) = C^2x^2 - D\sqrt{C}y - E(-\sqrt{C}x + \sqrt{A}y) + F,$$

which is the case when  $B = C = 0$ .

Now suppose  $\Delta^2 = B^2 - 4AC \neq 0$  and assume  $A \neq 0$ , then we have

$$f(x, y) = \left( \sqrt{A}x + \frac{B + \Delta}{2\sqrt{A}}y \right) \left( \sqrt{A}x + \frac{B - \Delta}{2\sqrt{A}}y \right) + Dx + Ey + F,$$

so we can apply the affine transformation

$$\begin{aligned} \sqrt{A}x + \frac{B + \Delta}{2\sqrt{A}}y &\mapsto x \\ \sqrt{A}x + \frac{B - \Delta}{2\sqrt{A}}y &\mapsto y \end{aligned}$$

which has determinant  $1/\Delta \neq 0$ , to obtain

$$f'(x, y) = xy + (\text{linear terms}).$$

Since affine coordinate rings are invariant up to affine transformations, we have either  $A(X) \simeq R$  or  $A(X) \simeq S$ . The ring  $R = k[x, y]/(y - x^2)$  is isomorphic to a polynomial ring over one variable, which is a factorial, hence it is integrally closed. The ring  $S = k[x, x^{-1}]$  is a discrete valuation ring, which are integrally closed by (A.M. p. 94).

- (b) The quadric surface  $Q_1$  can be covered by open affine varieties isomorphic to the affine surface  $z = xy$ . The affine coordinate ring of such surface is isomorphic to  $k[x, y]$ , which is integrally closed, hence  $Q_1$  is normal. Similarly,  $Q_2$  can be covered by open affine varieties isomorphic to either  $y = x^2$  or  $xy = 1$ , which was shown to be normal in (a).
- (c) The affine coordinate ring  $A(Y)$  is isomorphic to  $k[t^2, t^3]$ . The quotient field is  $k(t)$ , and the element  $t \in k(t)$  is integral over  $k[t^2, t^3]$  since  $(t)^2 - t^2 = 0$  but  $t \notin k[t^2, t^3]$ , hence  $A(Y)$  is not integrally closed, hence  $Y$  is not normal by (d).
- (d) Being integrally closed is a local property, that is a ring  $A$  is integrally closed if and only if  $A_{\mathfrak{m}}$  is integrally closed for all maximal ideals  $\mathfrak{m}$  in  $A$  by (A.M. p. 63). There is a one-to-one correspondence between maximal ideals of  $\mathcal{O}(Y) = A(Y)$  and points of  $Y$ , hence  $Y$  is normal if and only if  $\mathcal{O}_P$  is integrally closed for all  $P \in Y$  if and only if  $A(Y)_{\mathfrak{m}}$  is integrally closed for all maximal ideals  $\mathfrak{m}$  if and only if  $A(Y)$  is integrally closed.

- (e) We want to show there exists a normal scheme  $\widehat{Y}$  with morphism  $\pi : \widehat{Y} \rightarrow Y$  that is universal amongst all dominant morphisms from normal schemes into  $Y$ . Let  $A(Y)$  be the affine coordinate ring of  $Y$  and let  $B$  be the integral closure of  $A(Y)$  in its quotient field. Then,  $B$  is a finitely generated  $k$ -algebra by (3.9A), and it is a subring of a field, so it is an integral domain, which means it defines an affine variety. Let  $\widehat{Y}$  be an affine variety with coordinate ring  $A(\widehat{Y}) = B$ , and let  $\pi : \widehat{Y} \rightarrow Y$  be the morphism induced by the inclusion map  $A(Y) \hookrightarrow B$ . Let  $Z$  be a normal variety, and  $\varphi : Z \rightarrow Y$  a dominant morphism. Since  $Z$  can be covered by open affine varieties and local rings are preserved under open subsets, we reduce to the case when  $Z$  is an affine normal variety. We want to show there exists a ring homomorphism  $\alpha : \mathcal{O}(\widehat{Y}) \rightarrow \mathcal{O}(Z)$  such that the following diagram commutes:

$$\begin{array}{ccc} & \mathcal{O}(\widehat{Y}) & \\ \pi^* \nearrow & & \searrow \exists! \alpha \\ \mathcal{O}(Y) & \xrightarrow{\varphi^*} & \mathcal{O}(Z) \end{array}$$

If  $\varphi^* f = 0$  for some  $f \in \mathcal{O}(Y)$ , then  $f = 0$  on a closed dense subset of  $Y$ , hence  $f = 0$  on  $Y$ , so  $\varphi^*$  is injective. Thus,  $\varphi^*$  induces a map between the function fields  $\overline{\varphi^*} : K(Y) \rightarrow K(Z)$ , where we have the inclusions  $\mathcal{O}(Y) \hookrightarrow \mathcal{O}(\widehat{Y}) \hookrightarrow K(Y)$  by construction. If  $f \in \mathcal{O}(\widehat{Y})$ , then there exists an equation of integral dependence of the form

$$f^n + a_1 f^{n-1} + \cdots + a_n = 0, \quad a_i \in \mathcal{O}(Y),$$

and since  $\overline{\varphi^*}(a_i) = \varphi^*(a_i) \in \mathcal{O}(Z)$ , we can apply  $\overline{\varphi^*}$  to the equation above to obtain an equation of integral dependence of  $\overline{\varphi^*}(f)$  over  $\mathcal{O}(Z)$

$$\overline{\varphi^*}(f)^n + b_1 \overline{\varphi^*}(f)^{n-1} + \cdots + b_n = 0, \quad b_i = \varphi^*(a_i) \in \mathcal{O}(Z),$$

and since  $\mathcal{O}(Z)$  is integrally closed, we must have  $\overline{\varphi^*}(f) \in \mathcal{O}(Z)$ . Thus, we have  $\alpha = \overline{\varphi^*}|_{\mathcal{O}(\widehat{Y})}$  as the desired map, and it is unique by construction. Hence, we have a unique morphism  $\theta : Z \rightarrow \widehat{Y}$  such that  $\varphi = \pi \circ \theta$  and  $\theta^* = \alpha : \mathcal{O}(\widehat{Y}) \rightarrow \mathcal{O}(Z)$  by (3.5). □

**20.** Let  $Y$  be a variety of dimension  $\geq 2$ , and let  $P \in Y$  be a normal point. Let  $f$  be a regular function on  $Y - P$ .

- (a) Show that  $f$  extends to a regular function on  $Y$ .
- (b) Show this would be false for  $\dim Y = 1$ .

*Proof.*

- (a)  $Y$  can be covered by open affine varieties, so we reduce to the case when  $Y \subseteq \mathbb{A}^n$  itself is an affine variety. Then by composing with the inclusion morphism  $Y \rightarrow \mathbb{A}^n$ , we further reduce to the case when  $Y = \mathbb{A}^n$ , that is we want to show if  $f = g/h$  on  $\mathbb{A}^n - P$ , then we can extend  $f$  to a regular function on  $Y$ . We show that if a polynomial  $h \in k[x_1, \dots, x_n]$  is nonzero on  $\mathbb{A}^n - P$ , then  $f$  is nonzero on  $P$ . We can also assume  $P = (0, \dots, 0)$  and  $n = 2$  by induction. Suppose  $h(P) = 0$ , then  $h$  must have zero constant term, so we can write

$$h(x, y) = f_0(x)y^d + f_1(x)y^{d-1} + \cdots + f_d(x), \quad f_i(x) \in k[x] - k,$$

then there exists nonzero  $a \in k$  such that  $f_i(a) \neq 0$  for some  $i$ , we have

$$h(a, y) = a_0 y^d + a_1 y^{d-1} + \cdots + a_d \neq 0, \quad a_i = f_i(a),$$

and since  $k$  is algebraically closed,  $h(a, y)$  must have a solution  $y = b$ , so  $(a, b)$  is a solution to  $h$  that is not equal to  $P$ , a contradiction.

- (b) Consider  $Y = \mathbb{A}^1$  and  $P = 0$ , and let  $f = \frac{1}{x}$ . Then  $f$  is regular on  $Y - P$  by definition, but it clearly cannot be extended to the entire affine line. □